

Phase 2: Midterm Progress & Troubleshooting

(Confirm your submission items required from '2026 MEN310 Final Project.pdf'.)

0. Student Information

Name:

Student ID:

1. Initial Training & Loss Analysis

Instruction: Show your current Loss_Total, Loss_BC, and Loss_PDE curves. Analyze whether the loss is decreasing smoothly or showing instability.

(Insert Loss plots and analysis here...)

2. Ablation Study: Concept & Setup

****What is an Ablation Study?****

An 'Ablation Study' is a method to understand how different components of your AI model affect performance. By changing or 'removing' one variable at a time (like a specific layer, a type of activation function, or a set of data points), you can observe how that single change impacts the final result. It proves that your model's success isn't just luck, but the result of well-chosen components.

Example Comparison Points (Choose ONE):

- Activation Functions: Compare Tanh vs. ReLU. Observe performance for 2nd order PDEs.
- Point Density: Compare training with sparse vs. dense collocation points (e.g., 50 vs. 500).
- Network Capacity (Width & Depth): Compare different hidden layer structures (e.g., '2 layers x 16 nodes' vs. '4 layers x 64 nodes'). Observe differences in convergence speed and potential overfitting.

Instruction: Report the differences in result/convergence based on your chosen parameter.

(Write your Ablation Study setup and findings...)

3. Troubleshooting Experience (Flexible Reporting)

Instruction: Describe a significant challenge you encountered during model training and how you resolved it. Examples include (but are not limited to):

- Scaling issues (Exploding gradients requiring PDE normalization).
- Getting stuck in local minima.
- Autograd errors when implementing complex boundary conditions.

(Describe your specific problem and the applied solution...)